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Bake Mix (????? ????????)

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Commander Voice - ?????????? ?? .?

- 1 ?????????? ? ?????? ?????????????? ?????????? ?????? ? ?????????? ???
- 2 Multi-Chief: ??????????? ?????? ?????) ?????? ??????????? ???????
- 3 (?????? + ???) ??????? ??????? ?????????? ??????? ?? ?????? ??????? ??????????
- 4 Tactical Bottleneck AI ? Neural Load Radar ?????????? ??????????? ??? ??????? ??
- 5 IoT Predictive Maintenance ? Downtime Root-Cause Logging
- 6 (?????? ?????????? ??????? ??????????? ?????????? ?????????? ?? ??????) ??? ??????? ???????

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#	Module	#	Module
1	Predictive Flour Hydration AI	33	Multi-Feeder Guard
2	Dynamic Oven Thermal Planner	34	Rheon PMU Recall
3	Thermal Timeline Red/Amber	35	Thermal Friction Index
4	Live Energy & kWh Counter	36	Hydraulic Press Pressure & Thickness
5	Tactical Bottleneck AI	37	Multi-Deck & Rotary Oven Airflow Hydro-Balance
6	Neural Load Radar	38	Amperometric Torque Control
7	Cyber-Wave Audio Visualizer	39	Volumetric Divider & Piston Wear
8	IoT Predictive Maintenance	40	Dynamic Climate Control Proofer
9	Automated Yeast Maturity Tracker	41	Predictive Loading Bay & Route Dispatch
10	Cold Chain Auto-Recovery	42	Dynamic Rack & Trolley Buffer Tracking
11	Energy Tariff Auto-Shift	43	Synchronized Multi-Oven Unloading Wave
12	Voice Batch Traceability	44	Real-Time Ingredient Buffer Replenishment
13	Real-Time Yield & Waste Analytics	45	Executive Emergency Re-Routing AI
14	Multi-Zone Audio Routing	46	Dynamic Tray, Mold & Oven Tool Inventory
15	Dynamic Delivery Express Sync	47	Steam Recovery & Recirculation
16	Ghost Shift Auto-Pilot	48	Hands-Free Quick Maintenance Guide
17	Virtual Baker Coach	49	Downtime Root-Cause Logging
18	Digital Silo Telemetry	50	Multi-Deck Load Balancing AI
19	Equipment Stress Index	51	Startup Splash Screen (Glitch-Neon)
20	Voice Shift Handover Notes	52	Miki's Board
21	Offline Spatial Tour	53	Cyber-Industrial Kit (Space-Tech font)
22	Task Decomposition	54	Synchronized Success Sound
23	Offline Voice Glossary	55	Dynamic Focus Animation
24	Micro-Validation	56	Miki's Handshake (haptic)
25	Apprentice Progress Index	57	Mohamed's Lab Live View
26	Four-Hand Co-Weighing	58	Bake Mix - Interactive Training
27	SOS Colleague	59	Audio Snapshot
28	Voice Kudos	60	Smart Scale Sync
29	Shaping Metronome	61	Stress Index
30	Cross-Shift Handover Log	62	Recipe Scaling
31	Rheon Dough/Filling Ratio	63	Squad Check-In
32	Shutter Speed Sync	64	Multi-Chief Executive Architecture

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Milesight EM300-TH	Temperature + humidity LoRaWAN probe for cells and environment.	milesight.com
Efento (NB-IoT / BLE)	Temperature data logger for the cold chain (freezer/fridge).	getefento.com
Probes PT100 / DS18B20	Contact probes for dough core and oven (via gateway/ESP).	rs-online.com
Relay Shelly / Sonoff	WiFi relays to drive ovens, lights and cell heating.	shelly.com
Jabra wireless headset	Noise-cancelling headset for the hands-free Voice Core.	jabra.com

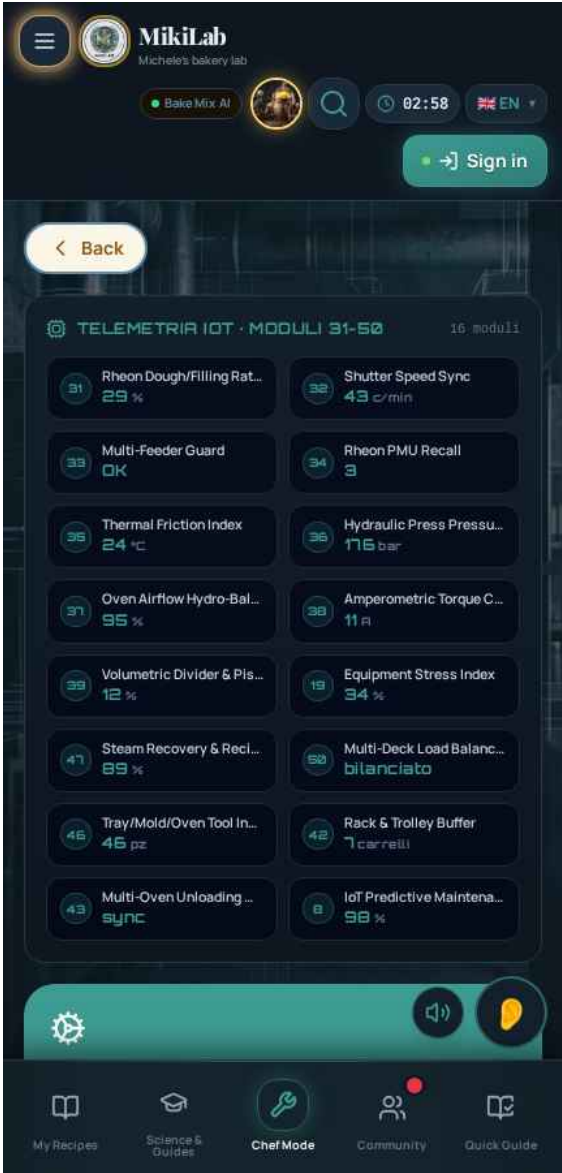
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Rheon lines (Encrusting/Extruder)	Dough/filling dosing and continuous forming. Modules 31-35.	Rheon
Hydraulic presses & molders	Pressure/thickness calibration with elasticity control. Module 36.	Bakery equipment vendors
Multi-deck & rotary ovens	Air/steam balance, steam recovery, load-balancing AI. Modules 37/47/50.	Wachtel, MIWE, Zucchelli
Spiral mixers & dividers	Amperometric torque, stress index, piston wear. Modules 38/19/39.	Pro mixer vendors
Proofing & freezing cells	Dynamic climate and cold-chain recovery. Modules 40/10.	Cell & blast-chiller vendors
IoT scales (Smart Scale)	Guided weighing and four-hand co-weighing, smart-scale sync. Modules 26/60.	Connected industrial scales
Industrial flour silos	Level and flour-flow telemetry. Module 18.	Silo & flour-transport vendors
Trays, molds, peels & tools	Cycle, cleaning and sanitizing tracking. Module 46.	Bakery tooling vendors

Talk with Miki - [redacted] [redacted] [redacted] [redacted] [redacted]
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IoT ????? - ?????



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